In addition to 20% randomly chosen data points from DDSE [15], additional 657 data points from other sources were considered for testing. Among these

ensure robustness, and external validation, ML model was validated against ~~a comprehensive set of~~ 1240 ~~ionic conductivity of solid electrolyte materials compiled from diverse sources~~. These included 583 ~~samples~~ from DDSE ~~in the range of 20 °C to 40 °C temperature,[15] and~~ 657 ~~additional samples~~ from external datasets not included in the training set; 442 from the University of Liverpool database in the range of 20 °C to 30 °C temperature,[16] is labelled as **A**, 175 from LLZO datasets at 27°C temperature,[17] is labelled as **B** and 40 from the dataset compiled by Sendek *et al.* at room temperature [7] is labelled as **C**.

data points

Samples

Do you need the subheading?

Model Development and Training : To address the wide range and skewed distribution of ionic conductivity (σ) values, ~~a logarithmic transformation~~ logarithm of conductivity ($\log_{10}\sigma$) was ~~applied~~ considered as the target variable, ~~This transformation helped to achieve~~ which helped in working with a ~~more uniform~~ (near-normal) distribution, as shown for ~~in~~ the compiled training dataset in Figure 2. Different regression models was trained that capable of predicting the ionic conductivity ($\log_{10}\sigma$) of solid electrolytes for all the available diverse data. The dataset was partitioned into training and testing subsets using an 80:20 split. ~~Three~~ Four distinct modelling experiments were conducted to systematically evaluate the effect of different feature sets (FS) on model performance:

Not correct. Additional 657 data points are there.

1. **Feature Set (FS -1)**: Training ~~using only the~~ based on 14 manually ~~engineered~~ selected features, ~~which were designed based on domain expertise to capture~~ relevant physicochemical ~~properties~~ by to the ionic conductivity.
2. **Feature Set (FS -2)**: Training ~~using only the~~ based on 200 Mat2Vec-generated features, which are vector representations of materials derived ~~from~~ through application of unsupervised machine learning techniques ~~trained~~ on materials science literature.
3. **Feature Set (FS -3)**: Training ~~using the~~ based on combined feature set of 214 features, comprising ~~both the manually engineered and the Mat2Vec features~~. 14 manually selected features (as in FS-1) and 200 Mat2 vect generated features (as in FS-2).

What about Feature Set FS-4 with 200+8?

4. Feature Set (FS -4):

Each model was evaluated on the test set to assess its predictive capability for $\log_{10}\sigma$. Several tree-based regression models were applied to predict ionic conductivity. The models considered ~~included~~ here are Decision Tree, Random Forest, Gradient Boosting, XGBoost, LightGBM, and Extra Trees Regressor. Their performance ~~was~~ were evaluated using the coefficient of determination (adjusted R^2 score - R^2_{adj}), which quantifies how well the models capture the variance in the experimental data. Several other error metrics were also computed. The respective equations are listed in supplementary document.

Error Metrics for Regression Analysis : ~~To statistically evaluate the prediction effectiveness of the regression model, several statistical error metrics were employed, as outlined in a previously published study.[18]~~ The adjusted coefficient of determination (R^2_{adj}) ~~was utilized to~~ helped in ascertaining the proportion of variation in the experimental data ~~that could be elucidated by~~ as per the model. This provided an overview of the model's fit quality. ~~We computed~~ The mean absolute error (MAE), and the root mean squared error (RMSE) ~~to assess~~ provided an assessment of accuracy in absolute terms. The RMSE exhibits heightened sensitivity to outliers as it penalizes greater deviations more severely. The mean bias error (MBE) was calculated to assess the model's persistent inclination to overpredict (positive bias) or underpredict (negative bias) relative to experimental observations. The standard error (SE of MBE)

Unlike FS-3, ~~manually selected features~~ ~~10~~ 10 physicochemical features were shortlisted from the 14 features ~~in FS-1~~ based on their significance in correlation metrics (to be discussed later). Total number features for training became 210.

Goes to Results & Discussion

Any reference to the equations in Appendix?

were computed in this study

what is the difference?

was subsequently employed to ascertain the statistical reliability of the bias. We also computed the standard deviation of errors (STD) to assess the dispersion and variability of the residuals across datasets. The mean target value (MTV) and mean predicted value (MPV) were presented to facilitate the comparison between the average conductivities predicted by the model and those measured in the experiment. Collectively, these several error measures constitute a comprehensive diagnostic framework: R^2_{adj} evaluates the model's fit to the data, while MAE and RMSE assess its accuracy and reliability. MBE and its standard error identify systematic errors, while STD indicates the variability of the data. This comprehensive review ensures that both the model's accuracy and bias are meticulously assessed. The dex ($\log_{10}\sigma$) scale illustrates conductivity and error values, with σ representing ionic conductivity in S cm^{-1}. A tenfold change (10^1) in σ corresponds to 1 dex. The extra file includes the equations utilized for the aforementioned error matrices.[18]

Goes to Results & Discussion

- **Results :** The ionic conductivity depends on various physico-chemical descriptors, which were explored in this study.

The results are organized to provide a clear understanding of the model development and its predictive reliability. First, the relevance and independence of the physical descriptors are evaluated, followed by the selection of an appropriate machine learning model. The subsequent sections present the quantitative error evaluation and a detailed diagnostic analysis of the model predictions to assess accuracy, bias, and uncertainty across different datasets.

Evaluation of Physical descriptors : The correlation matrix (Fig. 3) highlights how different physical and compositional descriptors relate to the logarithm of ionic conductivity across the dataset. Temperature ((T_K)) shows the strongest positive correlation (r ≈ 0.42), consistent with the thermally-activated nature of ionic transport in solid electrolytes (higher temperatures promote ion hopping and reduce activation barriers).[19] Among the composition-based descriptors, Li fraction (r ≈ 0.29) and Li-to-anion ratio (r ≈ 0.21) both correlate positively with conductivity. This reflects that lithium-rich compositions typically provide more mobile charge carriers and a greater probability of continuous conduction pathways a trend widely reported in garnet, NASICON and halide solid electrolytes.[20] A weaker but still positive correlation with packing efficiency proxy (r ≈ 0.23) suggests that moderately dense frameworks may optimize ion mobility by balancing structural openness with mechanical stability a concept supported by recent fast-ion conductor reviews which highlight that moderately rigid but not overly congested lattices favour high Li^+ mobility.[21] In contrast, several structural and electronic features show moderate negative correlations with conductivity. Average ionic radius (r ≈ -0.36) and heaviest element mass (r ≈ -0.36) both indicate that larger or heavier constituent ions hinder Li^+ migration likely by reducing lattice flexibility, increasing migration path length or raising phonon-mediated activation barriers.[22][23] Such mass/lattice-size effects are discussed in the recent literature on lattice dynamics and fast ionic transport.[22][23]

(new para)

with the proxy representing packing efficiency

(new para)

This is because Such constituent ions are likely to

this is consistent with the literature reports

why are these in "bold"?

As it put the electronegativity or its variance?

Goes to Results & Discussion

Similarly, **total atoms per formula unit** ($r \approx -0.32$) and **electronegativity variance** (($r \approx -0.29$)) show inverse relationships, implying that highly complex or chemically heterogeneous compositions may disrupt conduction pathways, or raise defect based scattering. ~~Reviews of structure-property correlations in solid electrolytes confirm~~ that simpler, more symmetric frameworks often outperform more complex chemistries. [24][25] A smaller negative correlation ~~is seen~~ for average electronegativity ($r \approx -0.26$) is consistent with the notion that stronger ionic bonding or higher electronegativity frameworks may localize Li-ions and reduce mobility. [26] The weak correlations observed for composition entropy, group diversity, and number of elements ($|r| < 0.20$) suggest that configurational complexity alone is not a strong predictor of transport properties when decoupled from specific structural or electronic effects. ~~Broad reviews on high-entropy and multi-component solid electrolytes emphasize that chemical complexity alone does not guarantee improved conductivity rather~~ the atomic connectivity and mobile-ion pathways matter most. [27][28]

(merge with previous paragraph)

What is this? Can you replace with chemical complexity?

alongside chemical complexity.

Overall, the trends confirm physically meaningful relationships: conductivity rises with temperature and lithium content, but decreases with increasing ion size, mass, bonding strength and structural complexity. These patterns are consistent with experimental studies of fast Li-ion conductors, reinforcing confidence that the descriptors capture genuine structure–property links rather than purely statistical artefacts.

and FS-4 considering reduced number of features alongside embedding features.

Machine learning model selection : As shown in Table 1, the coefficient of determination (R^2_{adj}) for different regression models is presented using ~~three~~ four types of training feature datasets: FS -1 with fourteen physical features, FS -2 with only embedding features, and FS -3 combining both physical and embedding features. The models trained on the combined feature set consistently outperformed those trained on either solely on physico-chemical descriptors the ~~domain-specific features~~ or the Mat2Vec embeddings alone. This suggests that the domain-specific features and the learned Mat2Vec embeddings capture complementary information, summarized in Table 1. The Random Forest Regressor achieved the highest R^2_{adj} value, demonstrating superior predictive accuracy relative to the other models. This ~~To further~~ refine ~~the physical descriptor set~~, correlation analysis was based on ~~performed to identify redundant features~~ treating the ones with next in the list were in figure — as redundant. Accordingly, i.e., ~~A~~ correlation threshold of 0.25 ~~was applied, and~~ six descriptors average atomic mass, number of elements, composition entropy, group diversity, Li-to-anion ratio, and lightest element mass were not considered (FS-4) ~~temporarily removed~~. However, ~~this reduction did not improve~~ the model performance (Table 2). Therefore, the complete set of fourteen physical descriptors was retained for further analysis here. ~~as it yielded the most accurate and consistent results (Table 1). Finally,~~ The Random Forest Regressor trained on the combined feature set (FS-3, containing both Mat2Vec and 14 physical descriptors) was selected for further evaluation and external validation.

Model Error Evaluation : To assess the dependability of the predictive model, various error measures were examined, including mean absolute error (MAE), root mean squared error (RMSE), and mean bias error (MBE). These indicators offer supplementary insights: MAE and RMSE

deteriorated with this refined ~~descrip~~ set of descriptors, suggesting the importance of all 14 physico-chemical descriptors in the mixed ~~the~~ basket. Also, an attempt was made to increase the pool of physico-chemical descriptors beyond the set of 14, and no improvement in prediction was observed.

To further explore the importance of physico-chemical ~~physical descriptors~~ features in the mixed basket of descriptors, the number of physicochemical descriptors was reduced to 108 from the initial set of 14.

discuss with me, this needs elaboration again in the text here.

(·) The

underestimates

quantify the average magnitude of prediction errors. MBE indicates if the model consistently ~~under-~~ or overestimates conductivity, and R^2_{adj} assesses the model's ability to represent experimental trends accurately. The model attained a R^2_{adj} of 0.85 and an RMSE of 0.56 dex for the 20% unseen test data of the DDSE dataset, as detailed in Table 3. Upon assessment using the external datasets shown in Table 4, the model exhibits robust predictive performance across all datasets, consistently achieving high R^2_{adj} values (≥ 0.84). Dataset B has the most accurate and fair forecasts. Datasets A and C, on the other hand, show systematic overprediction, which is shown by their positive bias ratings. The model has an overall RMSE of 0.72 dex and an MAE of 0.62 dex across 657 chemicals. This means that the average prediction errors are moderate, even when the model ~~is~~ was used on systems ~~that are chemically diverse~~ chemically diverse. ~~Despite some differences between the datasets, the study shows that the model works well in general and consistently finds conductivity patterns, with only a small bias that can be fixed in future updates.~~

bring Fig. S1 within the main manuscript.

Figure – presents the distribution of residual error. The plot appears to follow normal distribution with mean and standard deviation as — and — respectively. A normal distribution ~~suggests~~ suggests largely random error with no inherent bias, which in turn demonstrates the robustness of the predictive model.

The objective of this exercise is

Error Diagnostics and Residual analysis : The model's predictive reliability was evaluated through a comprehensive analysis of parity plots, residual distributions, regional error maps, and percentage error metrics to ascertain both global accuracy and local discrepancies across conductivity ranges, based on which the reliability of predictions can be reviewed. The parity plot (Fig. 4a) demonstrates a robust linear correlation between predicted and experimental ionic conductivities ($\log_{10} \sigma$), with the majority of data points situated near the ideal line (R^2_{adj} = 0.88). However, small upward deviations from overprediction appear in the low- and intermediate-conductivity range (< 10^{-5} S cm^{-1}), where the model slightly overestimates conductivity. These deviations ~~are~~ appeared consistently in ~~with~~ the residual analysis and regional error map (Figs. 4b and 4c). In particular, the datasets A and C show positive residuals and larger RMSE and MBE values (0.83 dex and 0.78 dex for A; 0.71 dex and 0.66 dex for C), while dataset B remains well-centred (RMSE = 0.29 dex, MBE ≈ 0 dex). The heat-map view of region-wise error further confirms that RMSE and MBE ~~are~~ were lowest for high-conductivity materials (>10^{-5} S cm^{-1}) and ~~rise~~ rose sharply below this threshold, reflecting limited data coverage and higher experimental uncertainty in that ~~regime~~ range.

These trends are supported by the Q–Q plots (Fig. 4d), which show that residuals for datasets A and B follow an ~~approximately~~ near-normal distribution, validating the use of RMSE and R^2_{adj} as representative performance metrics. Deviations from ~~normality~~ normal distribution appear ~~mainly in the tails~~ at the two ends of the plot, especially for dataset C, ~~again corresponding~~ This is due to a few outliers in the low-conductivity region. ~~This indicates that~~ The prediction errors ~~behave~~ varied randomly across most part of the range, with systematic bias confined to data-sparse zones. ~~where~~ This may be noted that 82.85% of compounds exhibited conductivities above 10^{-5} S cm^{-1} and only 17.14% ~~fall~~ fell below, as shown in figure 2. 17.15%

To provide a scale-independent perspective, absolute percentage errors (APEs) were examined (Fig. 4e). The overall mean absolute percentage error (MAPE ≈ 14 %) shows that predictions differ on average by about 14 % from experimental values. Among individual datasets, LLZO (~~B~~ Set B (Composite data)) shows the smallest median error (~6 % (space)), ~~Sendek (C) about~~ the median error was 11 %, and ~~Liverpool (A) about~~ the dataset A showed 17 %, consistent with their respective residual spreads. Only ~~a few points~~ (~11 out of 1253 data points) fall outside the ~~expected~~ whisker range, ~~confirming that the model's predictions are generally robust~~ suggesting a robust performance in general. These values are a median error of

Finally, residuals plotted against observed conductivities ~~(Fig. 4f) reveal that~~ where error variance ~~increases toward~~ lower σ values, reflecting both experimental uncertainty and model bias in data-scarce regions. Together, these analyses present a coherent picture: the model maintains strong predictive performance (R^2_{adj} = 0.88, RMSE = 0.72 dex, MAE = 0.63 dex, MBE = 0.56 dex, MAPE = 14 %), accurately preserves conductivity ranking, and remains well-calibrated for high-conductivity materials, while exhibiting moderate, interpretable overestimation in the low-conductivity regime.

For the dataset C (based Sendek et al.), Figure _(·) A greater

was observed for

■ Discussion :

No separate section required. Change the heading "Results" to "Results & Discussion".

Predictive reliability and uncertainty analysis: The results show that the machine learning model effectively captures the main trends in ionic conductivity across chemically diverse solid electrolytes, demonstrating strong generalization beyond the training data. Its ability to reproduce the relative ordering of conductivities makes it a practical tool for high-throughput screening of fast-ion conductors, particularly within the technologically relevant range of 10^{-5} to 10^{-2} S cm^{-1}. Minor deviations observed at the distribution tails suggest that extreme predictions may carry slightly higher uncertainty than those in the intermediate or higher conductivity range. However, these outliers reflect genuine physical variability rather than data noise and were therefore retained to preserve the true predictive behaviour of the model. The slight overprediction observed for low-conductivity materials mainly arises from data imbalance, emphasizing the need for more representative training datasets and uncertainty-aware modelling to ensure consistent accuracy across the entire conductivity spectrum. To correct this minor overestimation (quantified by MBE) and to further evaluate the model's statistical reliability, residuals from the unseen 20 % DDSE test data were analysed (Fig. S1). The residual distribution closely follows a normal curve ($\mu = 0.39$, $\sigma = 0.29$), confirming that the prediction errors are approximately normally distributed. This indicates that deviations between predicted and experimental conductivities are largely random and symmetric, without strong systematic bias. A similar residual pattern was also observed for the external dataset (Fig. 4b), further supporting this assumption and reinforcing the statistical soundness of the model's uncertainty estimation. The near-normal residual distribution further supports the use of RMSE-based confidence intervals to estimate predictive uncertainty. Accordingly, the uncertainty for any new prediction can be expressed as (predicted value – MBE) ± 1.96 × RMSE, corresponding to a 95 % confidence range under the assumption of normally distributed errors.

Model performance and feature representation: The current method works well because of both the regression framework and the way the features are represented. The Random Forest (RF) was always better than Gradient Boosting, XGBoost, LightGBM, Extra Trees, and single Decision Trees when they were all tested. Its group of de-correlated regression trees, along with bagging and random feature selection, lowers variance and stops overfitting. The RF model yields stable predictions with minimal tuning. Unlike boosting-based models, which can be sensitive to hyperparameter tuning and overfit noisy data. The built-in out-of-bag validation gives unbiased performance estimates, and the built-in feature-ranking ability makes it easy to deal with redundant or correlated variables.[29] Equally important is the choice of chemically informed descriptors. The integration of Mat2Vec embeddings with physically derived features enabled the model to encode both interpretable structural–property relationships and latent chemical knowledge, including periodic trends, bonding preferences, and electronegativity effects. Their effectiveness has been demonstrated in prior studies, where Mat2Vec descriptors combined with structural features in an

New para tab

The other point of interest in this study is the comparison of performances of different ML models.

True

Not ~~true~~

XGBoost framework enabled accurate classification of fast versus slow Li^+ conductors,[9] and in another case, were used to identify dopants in LLZO and classify materials likely to achieve both structural stability and high ionic conductivity.[11] In the present work, the use of 2,930 data points and a hybrid descriptor space (14 physical + 200 embedding features) provided sufficient diversity and depth to learn meaningful structure–conductivity relationships.

Broader implications and future improvements: These factors together show why the proposed framework works and how the combination of strong ensemble learning and chemically meaningful embeddings can lead to accurate, generalizable predictions of ionic conductivity. This method is a cost-effective way to help choose dopants or compounds, reduce trial-and-error in experimental synthesis, and speed up the search for high-performance solid electrolytes in a fraction of the time with good accuracy. Future research should investigate the integration of ensemble methods with graph neural networks, the expansion and rebalancing of datasets, and the incorporation of experimental uncertainty to further improve model reliability and transferability.

This discrepancy could be due to imbalance in available data for training.

However,

In any case,

Conclusion:

This study demonstrates that machine learning models can predict lithium ionic conductivity with ~~acceptable~~ reasonably good accuracy across diverse solid electrolyte datasets, ~~achieving strong overall performance~~ ($R^2 = 0.88$, RMSE = 0.72 dex). ~~Error diagnostics reveal that while~~ the predictions ~~are~~ were robust in the technologically relevant high-conductivity range (10^{-5} to 10^{-2} S cm^{-1}), (i.e.) ~~systematic~~ generally ed overestimation occurs in the low-conductivity ~~regime due to data imbalance~~ range. ~~Despite this limitation~~, the model reliably captures relative conductivity trends, making it a valuable tool for screening and prioritizing candidate materials. The near-normal residual distribution supports using RMSE-based confidence intervals for uncertainty estimation, where predictions can be expressed as *(predicted value – MBE) ± 1.96 × RMSE* within a 95% confidence range. Future improvements, including dataset rebalancing and bias correction strategies, will further enhance its ability to deliver accurate absolute predictions across the full conductivity spectrum. Overall, the synergy between robust ensemble learning and chemically informed embeddings provides an accurate, economical, and generalizable framework for predicting ionic conductivity and accelerating the discovery of high-performance solid electrolytes.

(New para)

Conclusions (added as a separate page)